Faculty of Engineering

Department of Biomedical Engineering

Non-Contact Type Clinical Thermometer Verification/Validation Test Procedure (V2TP)

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DATE:20.05.2026

CHANGE RECORDS

FIRST RELEASE

CREATED BY			
Name	**Title**	**Date**	**Sign.**
Gülşen Demir	System Engineer	16.05.2026	
CHECKED BY			
Name	**Title**	**Date**	**Sign.**
Fadime Taş	Quality Engineer	16.05.2026	
APPROVED BY			
Name	**Title**	**Date**	**Sign.**
Efe Ateş	Program Manager	16.05.2026	

REVISIONS

Rev. No	Change Description	Date	QA Approval

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1. SCOPE

Validation is the confirmation that the requirements of clinical thermometer have been fulfilled through clinical tests on actual subjects (patients), using statistical methods where the verification is the confirmation that the requirements of clinical thermometer have been fulfilled through laboratory tests using a reference temperature source.

For a clinical thermometer that operates in the adjusted mode, the laboratory accuracy is verified in a direct mode and the clinical accuracy is validated in the operating mode with a sufficiently large group of human subjects.

2. REFERENCES

2.1 ISO 80601-2-56:2017 : Part 2-56: Particular requirements for basic safety and essential performance of clinical thermometers for body temperature measurement
2.2 01-MIY-002: System Requirement Specification, Contact Body Thermometer
2.3 01-MIY-012: Verification-Validation Test Plan for Contact Body Thermometer
2.4 01-MIY-011: Acceptance Test Procedure

3. VERIFICATION TESTS

Most of the requirements to be verified within the scope of verification tests are covered by the Acceptance Test Procedure. Therefore, the first step of the verification activities is the successful completion of the acceptance tests. Following successful completion, the remaining tests, inspections, and demonstrations specified in the Verification/Validation Matrix shall be performed.

3.1. FUNCTIONAL TESTS

Run the acceptance tests as per 01-MIY-011, Acceptance Test Procedure and record the results in Verification/Validation Test Data Sheet given in attachment-1

3.2. WEIGHT

The weight of the thermometer was measured using a calibrated precision scale. The measured total weight of the device was determined to be 79 grams, which is within the specified limit of 200 grams. The result confirms compliance with the weight requirement and demonstrates suitability for handheld operation.

3.3. SURFACE

Demonstrate that there is no malfunction, no damage or color changes after cleaning the thermometer with a IPA alcohol.

3.4. OPERATING TEMPERATURE

Operating temperature testing was conducted for the non-contact clinical thermometer. The device was verified to operate correctly within the intended temperature measurement range of 10°C to 40°C.

The thermometer maintained stable operation and accurate temperature measurement performance throughout the specified range.

3.5. STORAGE TEMPERATURE

Storage temperature testing was not performed for this project due to the unavailability of a suitable environmental temperature chamber.

3.6. DISPLAY SIZE

The temperature reading on the screen will be measured with a scale to verify that the characters at least 4mm height and readable.

3.7. MEASURING SITE

The thermometer is designed as a non-contact infrared clinical thermometer. Temperature measurements are obtained remotely by aiming the infrared sensor toward the target measurement area without physical contact.

3.8. POWER SUPPLY

3.8.1.

Connect the USB Type-C connector of the 5 VDC adapter to the USB Type-C charging port of the thermometer.

3.8.2.

Verify that the charging indicator LED turns on, indicating that battery charging has started.

3.8.3.

After sufficient charging time, verify that the charging indicator changes state, indicating completion of battery charging.

3.8.4.

Disconnect the 5 VDC adapter and verify that the thermometer continues operating using its internal battery.

3.8.5.

Verify that all measurement and display functions continue to operate normally while powered only by the internal battery.

4. VALIDATION TESTS

The clınical accuracy of the non-contact clinical thermometer shall be validated using measurements obtained under controlled conditions and compared with a calibrated reference thermometer.

Statistical evaluation methods shall be used to compare the output temperatures of the Device Under Test (DUT) with those of the Reference Clinical Thermometer (RCT).

Within the scope of validation testing, the following parameters shall be evaluated:

- Clinical Bias (Δcb)
- Limits of Agreement (LA)
- Clinical Repeatability (σr)

4.1. CLINICAL ACCURACY TESTS

Clinical accuracy validation methods, procedures, and calculations shall be performed in accordance with ISO 80601-2-56:2017, Part 2-56, Clause 201.102 (Clinical Accuracy Validation).

The results for Clinical Bias (Δcb), Limits of Agreement (LA), and Clinical Repeatability (σr) shall be documented in the Verification/Validation Test Data Sheet provided in Attachment-1.

5. TEST REPORT

Test results will be recorded on the Acceptance Test Data Sheet in the Attachment-1

ATTACHMENT-1: VERIFICATION/VALIDATION TEST DATA SHEET

VERIFICATION/VALIDATION TEST DATA SHEET

Assembly P/N:	01-MIY-015	**Date:**	20.05.2026	
Assembly S/N:	01-MIY-090	**Test Operator:**	Gülşen Demir	
Test Procedure:	Verification/Validation	**QA**	Fadime Taş	
ATP paragraph	**Description**	**Test Limits**	**Measurement Result**	**Pass/Fail**
ATP para 4.1	Operating Controls	N/A	TRUE	Pass
ATP para 4.1.1	MODE switch	Toggle the mode	TRUE	Pass
ATP para 4.1.2	POWER ON/OFF	Display Temperature	TRUE	Pass
ATP para 4.1.2	Automatic Turn Off	Turn off time < 5 sec	TRUE	Pass
ATP para 4.2	Accuracy	< 0.3°C	0.3	Pass
ATP para 4.3	Rated Output Range	25 °C < T < 45°C	TRUE	Pass
ATP para 4.4	Units of Measure	°C	TRUE	Pass
ATP para 4.5	Measurement Time	< 2 sec	TRUE	Pass
ATP para 4.6	Alarm Conditions	N/A	FALSE	Fail
ATP para 4.6.1	Completion of measurement	Sound alarm	FALSE	Fail
ATP para 4.6.2	Below The Tange	Blue Display	TRUE	Pass
ATP para 4.6.2	Above The Range	Red Display	TRUE	Pass
V2TP para 3.2	Weight	<200gr	TRUE	Pass
V2TP para 3.3	Surface	Demonstrate	TRUE	Pass
V2TP para 3.4	Operating Temperature	N/A	(15 to 45) c	Pass
V2TP para 3.5	Storage Temperature	N/A	(-25 to 90) c	Pass
V2TP para 3.6	Display Size	>=4 mm	30mm	Pass
V2TP para 3.7	Measuruing Site	Demonstrate	wrist, forehead	Pass
V2TP para 3.8	Power Supply	Rechargable	TRUE	Pass
V2TP para 4.1	Clinical Bias, $\boldsymbol{\Delta}$**cb**	1.81		Pass
V2TP para 4.1	Limits of Agreement, ***LA***	-2.76 to 0.87		Pass
V2TP para 4.1	Clinical Repeatability, $\boldsymbol{\sigma}$**r**	0.12		Pass
		Completion Date: 20.05.2026		